

Status-Update-Slide-10-Speaker Notes

All of the comments EV left and I am working through them

Part 1

EV Comment: 1:51pm Jan 21 comment:

“could you elaborate what is a “behavioral building block” and the difference between that and the label in codebook? a concrete example would be helpful”

“similar to the comments on “cognitive function”: try to think a bit harder and use simpler language, what really are you trying to say? what is really the difference between those codes “blocks” or “cognitive functions”?”

in response to “annotation codes as basic behavioral building blocks, rather than the labels they were in the codebook.”

MX response:

What I was trying to get at with “behavioral building blocks” is the difference between describing what someone said and understanding what the behavior did for the idea over time.

In the codebook, each annotation (e.g. Knowledge Sharing, Evaluation Practices, Integration Practices) is a label for what type of behavior appears in a single utterance. At that level, the code is descriptive, but does not show how it contributes to the development of an idea across the meeting.

When I refer to the codes as “behavioral building blocks” or “cognitive functions” I am using the same annotations, but interpreting them by the role they play when combined with surrounding behaviors. This then focuses from individual utterances, to how ideas are constructed over time.

For e.g.

A Knowledge Sharing utterance such as:

- “Here’s what we found in the initial data.”
- This tells us information was introduced, but nothing about how it was used.

If this was followed by an Evaluation Practices utterance such as:

- “Does that scale with our constraints?”
- The Knowledge Sharing behavior now functions as the input that is being stress-tested, or evaluated.

If these are later followed by an Integration Practices utterance such as:

- “So we can frame this as X while accounting for Y”

Then those same codes collectively describe a sequence in which the information is introduced, evaluated, and synthesized. In this sense, the annotations act as building blocks for tracking how an idea develops, rather than as stand alone labels.

EV Comment 1:51PM Jan 21 comment:

“how is the cognitive function inferred ? how do you know what function each code serves?” “what will be the simplest language you can use to exchange the jargon here? so that everyone can understand what you meant”

in response to “I ended up grouping the original annotation codes by the cognitive function...”

MX response:

When I say that I grouped the annotation codes by “cognitive function”, I’m not saying that the function is explicitly encoded into the codebook nor it was inferred via some formal model. I had based it around the idea for each of the annotation codes from the codebook –

- *what does this behavior do to the set of ideas under discussion?*

This might mean asking whether it (1) adds/clarifies ideas, (2) questions, compares or contrasts ideas, (3) combine/reconciles ideas or even (4) does this lock in a direction OR move the group forward.

Based on that question:

- I treated Knowledge Sharing and Information Seeking as a way to expanding the idea space. This is b/c they introduce new information or clarify missing pieces of information.
- Evaluation Practices I had treated as a form of stress-testing. This is because they do not add any new information, but instead assess the possible feasibility/constraints/tradeoffs of an idea
- Integration Practices were treated as synthesis, because they explicitly combine or reconcile multiple ideas into a more coherent proposal.
- Coordination and Decision Practices were treated as commitment, because they signal alignment, next steps, or closure rather than exploration.

The “cognitive function” labels were not meant to introduce new theory. They were shorthand for how each behavior moves an idea forward when looking across multiple utterances, rather than at a single moment in isolation.

EV Comment 1:52PM Jan 21

“is this a cognitive function”?

in response to: “under a `Expansion/Seeding` category”

MX response:

When I used the term “cognitive function”, I wasn’t trying to claim a psychological function in

individuals. What I had meant was the functional role a behavior plays in the group's reasoning process.

E.g. I had labeled Knowledge Sharing and Information Seeking as "Expansion/Seeding" not b/c they represent a distinct cognitive function, but bc at the interaction level, they consistently expand the set of ideas or information under discussion. They increase what the team is considering, regardless of who is speaking or what mental process is involved.

EV Comment 1:58PM Jan 21 :

"what is your argument for this? for example, now i am evaluating your writing and leaving open questions and push back, do you not think this is expanding "idea space"? "wouldn't it also "expand" idea space by showing alternative idea spaces?"

in response to: "because it does not expand the idea space, but rather compares"

MX Response:

That's true when evaluation can sometimes expand idea space, especially when it exposes gaps, contradictions, or alternative decisions. I don't mean to claim that Evaluation Practices never expand ideas.

What I was trying to distinguish is the primary role the behavior plays at the moment it occurs. In my grouping, Expansion/Seeding behaviors directly introduce new candidate ideas or information into the discussion, whereas Evaluation Practices typically operate on ideas that are already on the table by questioning their feasibility, relevance, or fit.

Evaluation most often reshapes or constraints the existing idea space rather than adding new ideas outright, even though it can indirectly lead to new ideas later. That's why I grouped it separately as "stress-testing" rather than expansion.

EV Comment 2PM Jan 21:

"is this something you observed in the data? or is there literature to back it up (then add citations)"

in response to: "This had often marked a turning point between the exploration and convergence."

MX Response:

This claim is based on patterns observed in the SCIALOG data rather than directly from prior literature. In several meetings, Evaluation Practices tended to occur after idea expansion and before synthesis or coordination, coinciding with a shift toward narrowing the space of ideas. I treat this as an observed tendency in the dataset, not a universal rule.

Part 2

EV comment 2:17PM Jan 21

“even though i have read everything before this multiple times; i still am not sure why you chose to group codes into these categories: (1) what are your arguments for grouping them, (2) what goals are you trying to achieve and why you can't achieve this without these groups, and (3) how did you decide on these categories and which ones to be grouped? (purely based on your "idea" is not good enough an answer. could you provide evidence to support your argument. such as second order statistical analysis on relationships between codes, or quotes from data, or literature you could cite.)

could you try again to answer these questions specifically”

“i think you are getting there though in the sense that you are trying to separate codes by whether they are "expanding" or "contracting"; we just need more evidence and arguments to support your choices of grouping beyond just what you *think”

“and im just trying to push you little by little to be clearer with your thinking and arguments”

In response to “After I had defined all of these higher-level behavioral, ...”

MX Response:

Goal.

The goal of grouping the annotation codes was to test whether *higher-level behavioral patterns* over time could indicate idea convergence or narrowing in team meetings. Specifically, I was interested in whether meetings show directional movement from idea expansion toward consolidation or commitment, rather than simply documenting which behaviors occur.

Why grouping was necessary:

At the level of individual annotation codes, the data is too fine-grained and sparse to reason about trajectories across an entire meeting. Many codes appear infrequently, and analyzing transitions between all the original code results in a large number of low-signal transitions that are difficult to interpret in terms of convergence. Grouping the codes was a way to reduce complexity, and make it possible to examine whether families of related behaviors increase or decrease over time. This goal could not be achieved by treating each code independently, because convergence is inherently a higher-level process that unfolds across multiple utterances rather than within a single coded behavior.

How the groupings were chosen.

The groupings were initially hypothesis-driven and informed by two sources:

- (1) the functional definitions in the codebook, and
- (2) The recurring sequencing patterns observed in the SCIALOG transcripts

Rather than grouping arbitrarily, I grouped codes based on whether they tended to (a) introduce or clarify information (Knowledge Sharing, Information Seeking), (b) interrogate or constrain existing ideas (Evaluation Practices), (c) combine or reframe ideas (Integration Practices), or (d) formalize alignment and next steps (Coordination and Decision Practices). This reflects an expansion-contraction framing, where behaviors are differentiated by whether they tend to open the idea space or narrow it

Evidence and limitations:

This grouping is not presented as a definitive cognitive taxonomy. It is an exploratory operationalization intended to test a specific question about convergence dynamics. Evidence for this framing comes from observed temporal ordering in the data (e.g., evaluation frequently following knowledge sharing and preceding integration), as well as transition analyses showing non-random co-occurrence between certain codes. However, I acknowledge that stronger support would require second-order statistical analysis of code co-occurrence or explicit grounding in prior literature on idea generation and convergence. The limitations revealed by this approach ultimately motivated a shift toward linkography-based representations that preserve idea identity rather than relying solely on behavioral abstraction.

Part 3

EV Comment 2:10pm Jan 21

“show a concrete example (quotes) from your audit”

In response to: “I had figured out that these coordination and decision-like behaviors often increase near the end of the meetings due to the facilitation, reporting, or time management – not necessarily due to substantive agreement. I ended up running a false-positive diagnostic analysis while auditing the transcript for reference and using a lexical pattern detection. I then inspected these segments in context to distinguish when the meeting actually finished (because they ran out of time) compared to a genuine convergence, when people were wrapping up on an idea.”

MX Response:

In other words, I will need to find/select a single SCIALOG meeting where Coordination/Decision behaviors increase in the final third of meeting, BUT where the transcript shows the meeting ending due to time or reporting constraints rather than idea-level agreement.

- I will then include verbatim transcript excerpts illustrating (1) structural/logical closure and (2) a contrasting instance of genuine idea convergence, to make the distinction explicit

EV Comment 2:28PM Jan 21

“very interesting approach! makes sense!”

in response to: "using an entropy to measure this conceptual spread."

MX Response:

Noted, nothing else to do here.

EV Comment 2:29PM Jan 21

"this is super clear! love it." "more details on this method please: what is the unit of analysis and how did you calculate entropy

better with graphs and visuals"

in response to: "This motivation was because that the genuine convergence (e.g. a best idea, correct idea, final decision, successful outcome, etc.) should not only be reflected in how teams behave, but also to the extend the ideas are actually being considered."

MX Response:

To clarify the method:

The **unit of analysis** is a single SCIALOG meeting, split into temporal thirds (early vs. late).

For each third, I treat the distribution of ideas under discussion as a proxy for the team's conceptual state. Concretely, I operationalize this as the distribution of idea references (or idea states derived from transcript segmentation) within that segment.

Entropy is then computed over this distribution using standard Shannon entropy:

$$H = - \sum_i p_i \log p_i$$

where p_i represents the proportion of references to idea i within that temporal segment.

To assess narrowing, I compute entropy separately for the **first third** and **last third** of each meeting, and define a **pruning score** as:

$$\Delta H = H_{\text{late}} - H_{\text{early}}$$

Negative values indicate a reduction in conceptual spread (i.e., narrowing of the idea space).

In the slides, I will visualize this using:

- distributions of entropy values across meetings
- paired early vs. late entropy plots
- example meetings showing alignment vs. mismatch between behavioral convergence and entropy-based narrowing

EV Comment 2:32PM Jan 21

“could this test be used to support your decision in grouping codes based on their role in "expansion" vs "narrowing" of idea spaces? then you will have concrete evidence to support your decisions that will be much more convincing”

in response to: “computed the entropy separately for the first-third and last-third of each meeting. For that, I defined a pruning score as the change in entropy over time -- negative values indicate narrowing of an idea space.”

MX Response:

Yes - this was something I started to realize after computing the pruning score.

Because the entropy provides a code-agnostic signal of idea-space narrowing, it can be used as a validation layer for the behavioral groupings. The idea would be to test whether utterances coded as Expansion/Seeding tend to precede stable or increasing entropy, while Synthesis and Commitment behaviors tend to precede entropy decreases.

If that pattern holds consistently across meetings, it would provide empirical support that these grouped codes differ not just conceptually, but in their observable effect on the idea space. This would allow the grouping to be defended based on outcomes in the data, rather than solely on theoretical interpretation.

EV Comment 2:44 PM Jan 21

“how are you defining and measuring "convergence"?”

in response to: “`commitment/closure` reflected genuine idea-level convergence”

MX Response:

In this work, I define convergence operationally rather than normatively. I am not claiming that teams converge on the “best” or “correct” idea. Instead, I define convergence as a reduction in the set of ideas actively under discussion, accompanied by signals of alignment or forward movement.

I measured convergence along two complementary dimensions:

Behavioral convergence was inferred from increases in synthesis and commitment/coordination behaviors, particularly when these followed earlier exploratory and evaluative behaviors rather than occurring in isolation.

Conceptual convergence was measured using entropy over idea states, where a decrease in entropy from the first to the last third of a meeting indicates narrowing of the idea space.

I treat convergence as present when either or both of these signals appear, and I explicitly examine

cases where they diverge to avoid over-interpreting coordination or meeting closure as genuine idea-level convergence.

Part 4

Note to self: need to go through these remaining six comments.
(Jan 21 2026 at 4:55PM)

EV Comment 2:36PM Jan 21

“add graphs/visualizations to illustrate this. concrete examples from the data will also help. this is a very interesting finding”

for this part: “After this part was run, it was figured out that some of the meetings had strong `synthesis` and `commitment/closure` behaviors without a reduction in the entropy. Others showed conceptual narrowing without clear `commitment/closure` behavior. For this part I ended up checking against the transcript to figure out the exact cases and words spoken by the researchers.”

MX Comment:

Noted, will find examples + graph/viz.

EV Comment 2:39PM Jan 21

“examples from data please; or visualizations/graphs”

For this part: “this holds because teams can exhibit `synthesis` and coordination behaviors without adhering to a specific idea, and they can narrow their idea space without signaling commitment through decision-oriented behaviors.”

MX Comment:

Noted, will find examples + graph/viz.

EV Comment 2:39 Jan 21

“what is “transition dynamics”?”

For this part: “these annotations and transition dynamics can help figuring”

MX Comment:

By “transition dynamics”, I mean how teams move from one type of behavior to another over time, such as sharing information → evaluating it → synthesizing it. I focus on these shifts because they better reflect how ideas develop than any single annotation on its own.

EV Comment 3:05pm Jan 21

“why should this computation work? and *dumb this language down even more?”

For this part: “****to assess decision closure****, I computed late-stage gains in Commitment/Closure relative to early segments, and examined whether these gains occurred after synthesis rather than in isolation. ”

MX Comment:

The idea is that real decisions usually come after ideas have been worked through. By checking whether commitment increases near the end *and* follows synthesis, I’m testing whether closure looks like a natural outcome of reasoning rather than just a meeting ending.

EV Comment 3:07pm Jan 21

“why?”

For this part: “****To assess narrowing independently of behavioral annotations****, I computed idea-space entropy for the first and last thirds of each meeting, treating the distribution of idea states as a proxy for conceptual spread. ”

MX Comment:

The reason for using entropy here is that it provides a simple, code-agnostic way to quantify how spread out the ideas under discussion are at a given point in the meeting. When teams are exploring, conversation tends to be distributed across many different ideas. When teams narrow, discussion concentrates on fewer ideas more repeatedly.

EV Comment 3:07pm Jan 21

“why? what were you trying to achieve with this?”

For this part: “I compared behavioral convergence signals (`synthesis` and `commitment` gains) with entropy-based narrowing. ”

MX Comment:

I did this to check whether behaviors that signal convergence actually align with teams narrowing their ideas, or whether they sometimes occur without real conceptual agreement.